Experimental data for C (Carbon atom)

22 02 02 11 45

Other names
Carbon;

INChI	INChIKey	SMILES	IUPAC name
InChI=1S/C	OKTJSMMVPCPJKN-UHFFFAOYSA-N	[C]	

State	Conformation
$^3P_0$	KH

Enthalpy of formation (Hfg), Entropy, Integrated heat capacity (0 K to 298.15 K) (HH), Heat Capacity (Cp)

Property	Value	Uncertainty	units	Reference	Comment
Hfg(298.15K) $\Delta_f H^\circ$	716.68	0.45	kJ mol^{-1}	CODATA	
Hfg(0K) $\Delta_f H^\circ$	711.19	0.45	kJ mol^{-1}	CODATA	
Entropy (298.15K) S°	158.10	0.00	J K^{-1} mol^{-1}	CODATA	
Integrated Heat Capacity (0 to 298.15K) $H^\circ - H^\circ(T)$	6.54	0.00	kJ mol^{-1}	CODATA	
Heat Capacity (298.15K) C_p°	20.84		J K^{-1} mol^{-1}	Gurvich	

Information can also be found for this species in the NIST Chemistry Webbook

Electronic energy levels (cm^{-1})

Energy (cm^{-1})	Degeneracy	reference	description
0	1	NISTAtom	$^3P_0$
16.416713	3	NISTAtom	$^3P_1$
43.4134567	5	NISTAtom	$^3P_2$
10192.657	5	NISTAtom	^1D

Ionization Energies (eV)

Ionization Energy	I.E. unc.	vertical I.E.	v.I.E. unc.	reference
11.260				webbook

Electron Affinity (eV)

Electron Affinity	unc.	reference
1.262	0.000	webbook

Dipole, Quadrupole and Polarizability

Electric dipole moment

State	Config	State description	Conf description	Exp. min.	Dipole (Debye)				Reference	comment	Point Group	Components	
					x	y	z	total				dipole	quadrupole
1	1	$^3P_0$	K_h	True	0.000	0.000	0.000	0.000			K_h	0	0
2	1	$^3P_1$	K_h								K_h	0	0

3	1	$^3P_2$	K_h										
4	1	^1D	K_h	True									

Experimental dipole measurement abbreviations: MW microwave; DT Dielectric with Temperature variation; DR Indirect (usually an upper limit); MB Molecular beam

Calculated electric dipole moments for C (Carbon atom).

Electric quadrupole moment

State	Config	State description	Conf description	Exp. min.	Quadrupole (D Å)			Reference	comment	Point Group	Components	
					xx	yy	zz				dipole	quadrupole
1	1	$^3P_0$	K_h	True						K_h	0	0
2	1	$^3P_1$	K_h							K_h	0	0
3	1	$^3P_2$	K_h									
4	1	^1D	K_h	True								

Calculated electric quadrupole moments for C (Carbon atom).

Electric dipole polarizability (Å^3)

alpha	unc.	Reference
1.760	0.030	1977Mil/Bed:1

Calculated electric dipole polarizability for C (Carbon atom).

References

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squib	reference	DOI
1977Mil/Bed:1	TM Miller, B Bederson, "" Adv. At. Mol. Phys. 13, 1, 1977	10.1016/S0065-2199(08)60054-8
CODATA	Cox, J.D.; Wagman, D.D.; Medvedev, V.A.CODATA Key Values for Thermodynamics. Hemisphere, New York, 1989	10.1002/bbpc.19900940121
Gurvich	Gurvich, L.V.; Veyts, I. V.; Alcock, C. B., Thermodynamic Properties of Individual Substances, Fouth Edition, Hemisphere Pub. Co., New York, 1989	
NISTatom	NIST Atomic Spectra Database (http://physics.nist.gov/PhysRefData/ASD/levels_form.html)	10.18434/T4W30F
webbook	NIST Chemistry Webbook (http://webbook.nist.gov/chemistry)	10.18434/T4D303

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